

Book 2: Chapter 6 – Feature Importance Analysis

Machine Learning for Asset Managers (Marcos M. López de Prado)

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Chapter 6 Overview: Backtesting vs. Research Tool

Marcos' First Law of Backtesting

"Backtesting is not a research tool. Feature importance is." — Marcos López de Prado

The P-Hacking Trap

- Repeatedly tweaking models until a good backtest appears leads to **false discoveries**.
- At $\alpha = 0.05$, running 20 random iterations virtually guarantees a false positive.
- Backtests are **ex-post** simulations of monetization, not variable discovery tools.

The Role of Feature Importance

- Evaluates variable relevance **ex-ante**, before historical strategy simulation.
- Uncovers causal drivers and economic mechanisms behind financial phenomena.
- Robust to complex, high-dimensional, and non-linear interactions.

Topic 6.1: The Flaws of Classical p-Values

Classical Regression p-Values

P-values test $H_0 : \beta_j = 0$ under strict assumptions: correct specification, white-noise residuals, and un-multicollinear regressors.

Four Major Flaws in Finance

- 1 **Parametric Assumptions:** Unrealistic linear / normal distribution assumptions.
- 2 **Single Sample Variance:** High variance; no ensembling or variance reduction.
- 3 **Multicollinearity Failure:** Matrix inversion $(\mathbf{X}^T \mathbf{X})^{-1}$ explodes under correlation.
- 4 **In-Sample Only:** Fails to evaluate out-of-sample generalization.

Numerical Proof (40 Features)

- Dataset: 5 informative, 30 redundant, 5 noise features.
- Logit p -values deem only 4 non-noise features significant ($\alpha = 0.05$).
- Noise features rank higher than 14 true informative/redundant features due to substitution effects.

Topic 6.2: Mean Decrease Impurity (MDI)

In-Sample Explanatory Importance

Tree-based algorithm (e.g., Random Forest) that measures the weighted reduction in label impurity (entropy / Gini) at each split.

Mathematical Formulation

At node t , information gain from feature f is:

$$\Delta g[t, f] = i[t] - \frac{N_t^{(0)}}{N_t} i[t^{(0)}] - \frac{N_t^{(1)}}{N_t} i[t^{(1)}]$$

Averaged across M trees in an ensemble:

$$MDI_f = \frac{1}{M} \sum_{m=1}^M \sum_{t \in T_m, f(t)=f} \Delta g[t, f]$$

Bounded in $[0, 1]$ and $\sum_{f=1}^F MDI_f = 1$.

Properties & Limitations

- **Advantages:** Fast calculation; no distributional assumptions; ensemble bagging reduces variance; no matrix inversion.
- **Limitations:** In-sample (IS) only (does not test OOS prediction); diluted by substitution effects among correlated features.

Topic 6.3: Mean Decrease Accuracy (MDA)

Out-of-Sample Predictive Importance

Classifier-agnostic method that measures the decay in cross-validated performance when shuffling feature values.

Algorithm Procedure

- 1 Fit model on training set; evaluate baseline OOS score S_{OOS} on purged/embargoed test fold.
- 2 Permute/shuffle values of feature f in test fold.
- 3 Re-evaluate OOS score $S_{\text{OOS}}^{(f)}$.
- 4 Compute importance decay:

$$MDA_f = \frac{S_{\text{OOS}} - S_{\text{OOS}}^{(f)}}{S_{\text{OOS}}}$$

Properties & Limitations

- **Advantages:** Out-of-sample (OOS) cross-validated; works on any classifier; can yield negative values for harmful features.
- **Limitations:** Computationally expensive; susceptible to substitution effects (correlated twin features mask each other).

Topic 6.4: Scoring Metrics – Probability-Weighted Accuracy

The Problem with Raw Accuracy in Finance

Accuracy scores correct binary calls without accounting for prediction probability. A classifier making good calls with low confidence and bad calls with high confidence can still achieve high accuracy, leading to severe financial drawdowns.

Log-Loss / Cross-Entropy Loss

Scores the average log-likelihood of true labels:

$$\text{Log-Loss} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]$$

Heavily penalizes confident incorrect predictions.

Probability-Weighted Accuracy (PWA)

Negative Average Likelihood (NegAL):

$$\text{NegAL} = -\text{Log-Loss}$$

Provides an interpretable, confidence-calibrated scoring function for financial MDA permutation analysis.

Topic 6.5: Substitution Effects & Orthogonalization

The Substitution Effect (Multicollinearity)

When features share predictive information, MDI dilutes importance across substitutes, while MDA marks both twin features as unimportant because shuffling one is compensated by the other.

PCA Orthogonalization

- Transform features $\mathbf{X}$ into principal components $\mathbf{P} = \mathbf{XW}$.
- Run MDI/MDA on orthogonal components $\mathbf{P}$.
- **Three Caveats:** Non-linear substitution fails; PCs lose economic interpretability; eigenvectors maximize variance without regarding labels.

Confirmatory Proof against Overfitting

- Unsupervised PCA ranks components without label information.
- High correlation (Kendall's $\tau \approx 0.81$) between PCA eigenvalues and MDI scores confirms non-overfit feature discovery.

Topic 6.6: Clustered Feature Importance (CFI)

Taming Multicollinearity without Basis Change

Clustered Feature Importance (CFI) clusters similar features into mutually dissimilar groups and performs MDI/MDA at the cluster level, preserving original feature definitions.

Step 1: Feature Clustering

- Partition features into K clusters $C_1, \dots, C_K$ using Optimal Number of Clusters (ONC) on correlation or Variation of Information.
- *Optional Residualization*: Replace X_i with residual $\hat{\epsilon}_i$ to remove cross-cluster overlap if silhouette scores are weak.

Step 2: Clustered Importance

- **Clustered MDI**: Sum MDI scores across all features in cluster C_k :

$$MDI_{C_k} = \sum_{f \in C_k} MDI_f$$

- **Clustered MDA**: Simultaneously shuffle all features in cluster C_k and measure OOS score decay.

Topic 6.7: Empirical Proof – Standard vs. Clustered Importance

40-Feature Benchmark Experiment (5 Informative, 30 Redundant, 5 Noise)

Comparing standard MDI/MDA against Clustered Feature Importance (CFI).

Standard MDI / MDA Failure

- Substitution effects dilute MDI/MDA scores of redundant features.
- High variance causes some noise features to rank close to informative variables.
- Fails to reflect equal underlying economic importance.

Clustered MDI / MDA (CFI) Success

- ONC correctly recovers 6 clusters (5 informative clusters + 1 noise cluster C_5).
- Noise cluster C_5 importance drops to ≈ 0 .
- All 5 informative clusters $C_0, \dots, C_4$ achieve equal, high importance (≈ 0.20 each).

Summary: Key Takeaways on Feature Importance

Core Principles for Quantitative Asset Managers

- ① **Backtesting vs. Research:** Use feature importance ex-ante as a true research tool to discover causal economic theories before running backtests.
- ② **Abandon p-Values:** Classical regression p -values fail in finance due to multicollinearity, matrix inversion instability, and lack of OOS cross-validation.
- ③ **Combine MDI and MDA:** MDI provides fast, in-sample tree explanatory power; MDA provides purged, cross-validated out-of-sample predictive power.
- ④ **Leverage Clustered Feature Importance (CFI):** Combine ONC clustering with MDI/MDA to completely neutralize substitution effects and multicollinearity.

Thank You!

Questions & Discussion

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